Data-Driven Emulation of Modal Aerosol Microphysics via Neural Operator-Based Modeling

BERKELEY LAB

Scientific Achievement

- Developed ADON (Aerosol Deep Operator Network), a physics-informed AI model that reproduces complex aerosol microphysics in DOE's Energy Exascale Earth System Model (E3SMv2).
- Trained on 9.8 million samples from a global ~1° resolution simulation, ADON achieved over 95.7% accuracy (R^2) in predicting aerosol behavior — even in scenarios beyond its training data.
- Demonstrated that ADON can generalize to new forecasting situations while maintaining high fidelity compared to traditional simulations.

Significance and Impact

- Accelerates atmospheric simulations by replacing the most computationally expensive aerosol calculations with a fast, AI-based surrogate.
- Reduces computing costs and energy use while enabling higher-resolution, longer-term forecasts.
- Provides a generalizable framework for integrating AI with physics-based models — applicable to many areas of Earth system science.

Technical Approach

- Uses a dual-network architecture: one network learns aerosol–atmosphere relationships, the other encodes spatial grids, time, and physics-based patterns extracted from data.
- Incorporates feature importance analysis to identify which physical variables most influence predictions.
- Runs efficiently on GPU nodes, including reduced-precision computing for even faster performance.

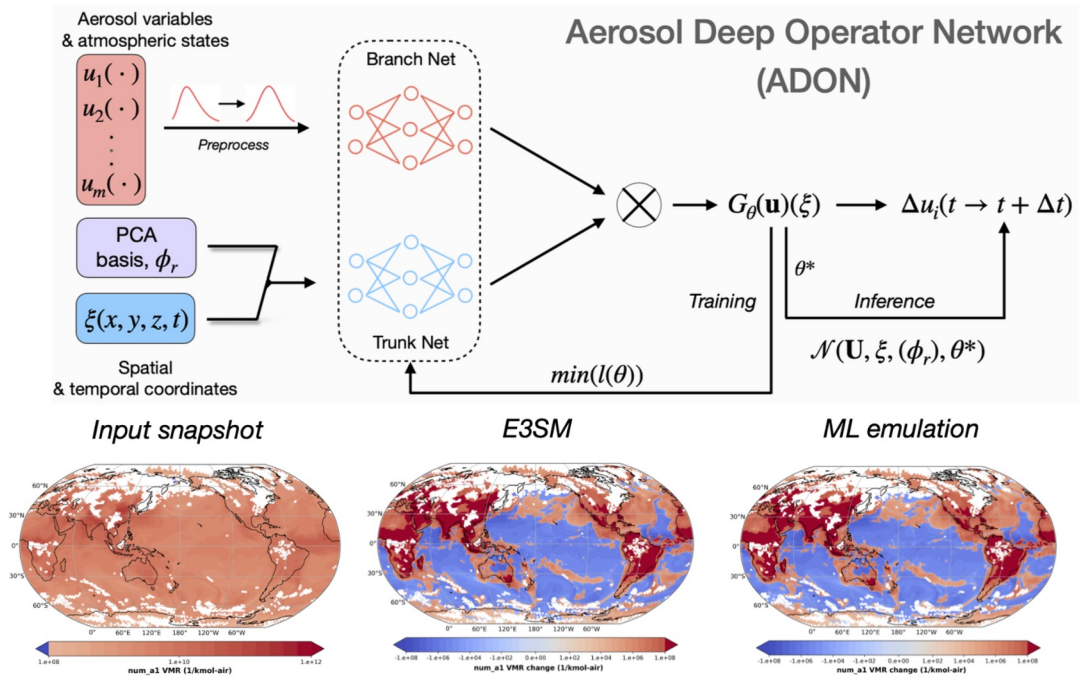

FIG: (Top) Schematic illustration of the aerosol deep operator network (ADON). The dual-net architecture is used in both training and inference to learn the solution operator for aerosol microphysics. (Bottom) Global view of the VMR tendency of key aerosol components over a 1800-second interval, as predicted by E3SM (middle) and by our machine-learning emulator (right), shown for the lowest model layer near the Earth's surface.

PI(s)/Facility Lead(s): Ann Almgren, Leonid Oliker, Hui Wan
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ASCR PM: Xujing Davis, Marco Fornari
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